



MATH120: Derivatives Rules

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Learning Objectives

- Apply the product rule to differentiate products of functions
- Apply the quotient rule to differentiate rational functions
- Apply the chain rule to differentiate composite functions
- Combine differentiation rules for complex expressions

Simplify each expression completely. Show all steps and circle your final answer.

Differentiating polynomials

1. A company's total revenue is $R(x) = 1x^3 + -3x^2 + 13x + 15$. Find the marginal revenue $R'(1)$.

$$f(x) = 1x^3 + -3x^2 + 13x + 15, x = 1$$

Answer: _____

2. A company's total revenue is $R(x) = 1x^3 + 0x^2 + 6x + 37$. Find the marginal revenue $R'(3)$.

$$f(x) = 1x^3 + 0x^2 + 6x + 37, x = 3$$

Answer: _____

3. A company's total revenue is $R(x) = 2x^3 + -1x^2 + 16x + 40$. Find the marginal revenue $R'(1)$.

$$f(x) = 2x^3 + -1x^2 + 16x + 40, x = 1$$

Answer: _____

4. A company's total revenue is $R(x) = 1x^3 + 0x^2 + 14x + 30$. Find the marginal revenue $R'(2)$.

$$f(x) = 1x^3 + 0x^2 + 14x + 30, x = 2$$

Answer: _____

5. A company's total revenue is $R(x) = 1x^3 + -2x^2 + 11x + 2$. Find the marginal revenue $R'(1)$.

$$f(x) = 1x^3 + -2x^2 + 11x + 2, x = 1$$

Answer: _____

6. A company's total revenue is $R(x) = 1x^3 + -3x^2 + 12x + 29$. Find the marginal revenue $R'(2)$.

$$f(x) = 1x^3 + -3x^2 + 12x + 29, x = 2$$

Answer: _____



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7. A company's total revenue is $R(x) = 1x^3 + -1x^2 + 18x + 15$. Find the marginal revenue $R'(2)$.

$$f(x) = 1x^3 + -1x^2 + 18x + 15, x = 2$$

Answer: _____

8. A company's total revenue is $R(x) = 1x^3 + -1x^2 + 19x + 28$. Find the marginal revenue $R'(1)$.

$$f(x) = 1x^3 + -1x^2 + 19x + 28, x = 1$$

Answer: _____

Power rule

9. Use the power rule to find the derivative of $f(x) = 1x^3$. Then evaluate $f'(3)$.

$$f(x) = 1x^3, x = 3$$

Answer: _____

10. Find the instantaneous rate of change of $C(x) = 5x^4$ at $x = 1$. (Use the power rule.)

$$f(x) = 5x^4, x = 1$$

Answer: _____

11. Use the power rule to find the derivative of $f(x) = 3x^2$. Then evaluate $f'(2)$.

$$f(x) = 3x^2, x = 2$$

Answer: _____

12. Find the instantaneous rate of change of $C(x) = 6x^3$ at $x = 2$. (Use the power rule.)

$$f(x) = 6x^3, x = 2$$

Answer: _____

13. Use the power rule to find the derivative of $f(x) = 2x^4$. Then evaluate $f'(3)$.

$$f(x) = 2x^4, x = 3$$

Answer: _____

14. Find the instantaneous rate of change of $C(x) = 4x^2$ at $x = 4$. (Use the power rule.)

$$f(x) = 4x^2, x = 4$$

Answer: _____



15. Use the power rule to find the derivative of $f(x) = 1x^2$. Then evaluate $f'(1)$.

$$f(x) = 1x^2, \quad x = 1$$

Answer: _____

16. Find the instantaneous rate of change of $C(x) = 8x^4$ at $x = 1$. (Use the power rule.)

$$f(x) = 8x^4, \quad x = 1$$

Answer: _____

17. Use the power rule to find the derivative of $f(x) = 2x^2$. Then evaluate $f'(3)$.

$$f(x) = 2x^2, \quad x = 3$$

Answer: _____

18. Find the instantaneous rate of change of $C(x) = 4x^3$ at $x = 2$. (Use the power rule.)

$$f(x) = 4x^3, \quad x = 2$$

Answer: _____

19. Use the power rule to find the derivative of $f(x) = 4x^3$. Then evaluate $f'(2)$.

$$f(x) = 4x^3, \quad x = 2$$

Answer: _____

20. Find the instantaneous rate of change of $C(x) = 5x^2$ at $x = 4$. (Use the power rule.)

$$f(x) = 5x^2, \quad x = 4$$

Answer: _____

21. Use the power rule to find the derivative of $f(x) = 1x^2$. Then evaluate $f'(2)$.

$$f(x) = 1x^2, \quad x = 2$$

Answer: _____

22. Find the instantaneous rate of change of $C(x) = 7x^3$ at $x = 1$. (Use the power rule.)

$$f(x) = 7x^3, \quad x = 1$$

Answer: _____

23. Use the power rule to find the derivative of $f(x) = 3x^4$. Then evaluate $f'(3)$.

$$f(x) = 3x^4, \quad x = 3$$

Answer: _____



Product rule

24. Use the product rule to differentiate $f(x) = (3x + 4)(4x + 3)$. Find $f'(0)$.

$$f(x) = (3x + 4)(4x + 3), x = 0$$

Answer: _____

25. Use the product rule to differentiate $f(x) = (1x + 3)(4x + 1)$. Find $f'(2)$.

$$f(x) = (1x + 3)(4x + 1), x = 2$$

Answer: _____

26. Use the product rule to differentiate $f(x) = (2x + 4)(4x + 6)$. Find $f'(2)$.

$$f(x) = (2x + 4)(4x + 6), x = 2$$

Answer: _____

27. Use the product rule to differentiate $f(x) = (1x + 3)(1x + 2)$. Find $f'(2)$.

$$f(x) = (1x + 3)(1x + 2), x = 2$$

Answer: _____

28. Use the product rule to differentiate $f(x) = (4x + 2)(2x + 3)$. Find $f'(2)$.

$$f(x) = (4x + 2)(2x + 3), x = 2$$

Answer: _____

29. Use the product rule to differentiate $f(x) = (4x + 3)(3x + 5)$. Find $f'(1)$.

$$f(x) = (4x + 3)(3x + 5), x = 1$$

Answer: _____

30. Use the product rule to differentiate $f(x) = (1x + 1)(2x + 1)$. Find $f'(2)$.

$$f(x) = (1x + 1)(2x + 1), x = 2$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Differentiating polynomials, Product rule, Power rule. All answers verified by independent computation.

Solutions



Differentiating polynomials

1. A company's total revenue is $R(x) = 1x^3 + -3x^2 + 13x + 15$. Find the marginal revenue $R'(1)$.

$$f(x) = 1x^3 + -3x^2 + 13x + 15, x = 1$$

$$\rightarrow R'(x) = 3x^2 + -6x + 13.$$

$$\rightarrow R'(1) = 3(1)^2 + -6(1) + 13 = 10.$$

$\rightarrow$ Marginal revenue at $x = 1$ units is \$10.

Answer: $f'(x) = 3x^2 + -6x + 13, f'(1) = 10$

2. A company's total revenue is $R(x) = 1x^3 + 0x^2 + 6x + 37$. Find the marginal revenue $R'(3)$.

$$f(x) = 1x^3 + 0x^2 + 6x + 37, x = 3$$

$$\rightarrow R'(x) = 3x^2 + 0x + 6.$$

$$\rightarrow R'(3) = 3(3)^2 + 0(3) + 6 = 33.$$

$\rightarrow$ Marginal revenue at $x = 3$ units is \$33.

Answer: $f'(x) = 3x^2 + 0x + 6, f'(3) = 33$

3. A company's total revenue is $R(x) = 2x^3 + -1x^2 + 16x + 40$. Find the marginal revenue $R'(1)$.

$$f(x) = 2x^3 + -1x^2 + 16x + 40, x = 1$$

$$\rightarrow R'(x) = 6x^2 + -2x + 16.$$

$$\rightarrow R'(1) = 6(1)^2 + -2(1) + 16 = 20.$$

$\rightarrow$ Marginal revenue at $x = 1$ units is \$20.

Answer: $f'(x) = 6x^2 + -2x + 16, f'(1) = 20$

4. A company's total revenue is $R(x) = 1x^3 + 0x^2 + 14x + 30$. Find the marginal revenue $R'(2)$.

$$f(x) = 1x^3 + 0x^2 + 14x + 30, x = 2$$

$$\rightarrow R'(x) = 3x^2 + 0x + 14.$$

$$\rightarrow R'(2) = 3(2)^2 + 0(2) + 14 = 26.$$

$\rightarrow$ Marginal revenue at $x = 2$ units is \$26.

Answer: $f'(x) = 3x^2 + 0x + 14, f'(2) = 26$

5. A company's total revenue is $R(x) = 1x^3 + -2x^2 + 11x + 2$. Find the marginal revenue $R'(1)$.

$$f(x) = 1x^3 + -2x^2 + 11x + 2, x = 1$$

$$\rightarrow R'(x) = 3x^2 + -4x + 11.$$

$$\rightarrow R'(1) = 3(1)^2 + -4(1) + 11 = 10.$$

$\rightarrow$ Marginal revenue at $x = 1$ units is \$10.

Answer: $f'(x) = 3x^2 + -4x + 11, f'(1) = 10$



6. A company's total revenue is $R(x) = 1x^3 + -3x^2 + 12x + 29$. Find the marginal revenue $R'(2)$.

$$f(x) = 1x^3 + -3x^2 + 12x + 29, x = 2$$

$$\rightarrow R'(x) = 3x^2 + -6x + 12.$$

$$\rightarrow R'(2) = 3(2)^2 + -6(2) + 12 = 12.$$

$\rightarrow$ Marginal revenue at $x = 2$ units is \$12.

Answer: $f'(x) = 3x^2 + -6x + 12, f'(2) = 12$

7. A company's total revenue is $R(x) = 1x^3 + -1x^2 + 18x + 15$. Find the marginal revenue $R'(2)$.

$$f(x) = 1x^3 + -1x^2 + 18x + 15, x = 2$$

$$\rightarrow R'(x) = 3x^2 + -2x + 18.$$

$$\rightarrow R'(2) = 3(2)^2 + -2(2) + 18 = 26.$$

$\rightarrow$ Marginal revenue at $x = 2$ units is \$26.

Answer: $f'(x) = 3x^2 + -2x + 18, f'(2) = 26$

8. A company's total revenue is $R(x) = 1x^3 + -1x^2 + 19x + 28$. Find the marginal revenue $R'(1)$.

$$f(x) = 1x^3 + -1x^2 + 19x + 28, x = 1$$

$$\rightarrow R'(x) = 3x^2 + -2x + 19.$$

$$\rightarrow R'(1) = 3(1)^2 + -2(1) + 19 = 20.$$

$\rightarrow$ Marginal revenue at $x = 1$ units is \$20.

Answer: $f'(x) = 3x^2 + -2x + 19, f'(1) = 20$



Power rule

9. Use the power rule to find the derivative of $f(x) = 1x^3$. Then evaluate $f'(3)$.

$$f(x) = 1x^3, \quad x = 3$$

$$\rightarrow \text{Power rule: } d/dx[1x^3] = 3x^2.$$

$$\rightarrow f'(3) = 3(3)^2 = 27.$$

Answer: $f'(x) = 3x^2, \quad f'(3) = 27$

10. Find the instantaneous rate of change of $C(x) = 5x^4$ at $x = 1$. (Use the power rule.)

$$f(x) = 5x^4, \quad x = 1$$

$$\rightarrow C'(x) = 20x^3.$$

$$\rightarrow C'(1) = 20(1)^3 = 20.$$

Answer: $f'(x) = 20x^3, \quad f'(1) = 20$

11. Use the power rule to find the derivative of $f(x) = 3x^2$. Then evaluate $f'(2)$.

$$f(x) = 3x^2, \quad x = 2$$

$$\rightarrow \text{Power rule: } d/dx[3x^2] = 6x^1.$$

$$\rightarrow f'(2) = 6(2)^1 = 12.$$

Answer: $f'(x) = 6x^1, \quad f'(2) = 12$

12. Find the instantaneous rate of change of $C(x) = 6x^3$ at $x = 2$. (Use the power rule.)

$$f(x) = 6x^3, \quad x = 2$$

$$\rightarrow C'(x) = 18x^2.$$

$$\rightarrow C'(2) = 18(2)^2 = 72.$$

Answer: $f'(x) = 18x^2, \quad f'(2) = 72$

13. Use the power rule to find the derivative of $f(x) = 2x^4$. Then evaluate $f'(3)$.

$$f(x) = 2x^4, \quad x = 3$$

$$\rightarrow \text{Power rule: } d/dx[2x^4] = 8x^3.$$

$$\rightarrow f'(3) = 8(3)^3 = 216.$$

Answer: $f'(x) = 8x^3, \quad f'(3) = 216$

14. Find the instantaneous rate of change of $C(x) = 4x^2$ at $x = 4$. (Use the power rule.)

$$f(x) = 4x^2, \quad x = 4$$

$$\rightarrow C'(x) = 8x^1.$$

$$\rightarrow C'(4) = 8(4)^1 = 32.$$

Answer: $f'(x) = 8x^1, \quad f'(4) = 32$



15. Use the power rule to find the derivative of $f(x) = 1x^2$. Then evaluate $f'(1)$.

$$f(x) = 1x^2, \quad x = 1$$

$$\rightarrow \text{Power rule: } d/dx[1x^2] = 2x^1.$$

$$\rightarrow f'(1) = 2(1)^1 = 2.$$

Answer: $f'(x) = 2x^1, \quad f'(1) = 2$

16. Find the instantaneous rate of change of $C(x) = 8x^4$ at $x = 1$. (Use the power rule.)

$$f(x) = 8x^4, \quad x = 1$$

$$\rightarrow C'(x) = 32x^3.$$

$$\rightarrow C'(1) = 32(1)^3 = 32.$$

Answer: $f'(x) = 32x^3, \quad f'(1) = 32$

17. Use the power rule to find the derivative of $f(x) = 2x^2$. Then evaluate $f'(3)$.

$$f(x) = 2x^2, \quad x = 3$$

$$\rightarrow \text{Power rule: } d/dx[2x^2] = 4x^1.$$

$$\rightarrow f'(3) = 4(3)^1 = 12.$$

Answer: $f'(x) = 4x^1, \quad f'(3) = 12$

18. Find the instantaneous rate of change of $C(x) = 4x^3$ at $x = 2$. (Use the power rule.)

$$f(x) = 4x^3, \quad x = 2$$

$$\rightarrow C'(x) = 12x^2.$$

$$\rightarrow C'(2) = 12(2)^2 = 48.$$

Answer: $f'(x) = 12x^2, \quad f'(2) = 48$

19. Use the power rule to find the derivative of $f(x) = 4x^3$. Then evaluate $f'(2)$.

$$f(x) = 4x^3, \quad x = 2$$

$$\rightarrow \text{Power rule: } d/dx[4x^3] = 12x^2.$$

$$\rightarrow f'(2) = 12(2)^2 = 48.$$

Answer: $f'(x) = 12x^2, \quad f'(2) = 48$

20. Find the instantaneous rate of change of $C(x) = 5x^2$ at $x = 4$. (Use the power rule.)

$$f(x) = 5x^2, \quad x = 4$$

$$\rightarrow C'(x) = 10x^1.$$

$$\rightarrow C'(4) = 10(4)^1 = 40.$$

Answer: $f'(x) = 10x^1, \quad f'(4) = 40$



21. Use the power rule to find the derivative of $f(x) = 1x^2$. Then evaluate $f'(2)$.

$$f(x) = 1x^2, \quad x = 2$$

→ Power rule: $d/dx[1x^2] = 2x^1$.

$$\rightarrow f'(2) = 2(2)^1 = 4.$$

Answer: $f'(x) = 2x^1, \quad f'(2) = 4$

22. Find the instantaneous rate of change of $C(x) = 7x^3$ at $x = 1$. (Use the power rule.)

$$f(x) = 7x^3, \quad x = 1$$

→ $C'(x) = 21x^2$.

$$\rightarrow C'(1) = 21(1)^2 = 21.$$

Answer: $f'(x) = 21x^2, \quad f'(1) = 21$

23. Use the power rule to find the derivative of $f(x) = 3x^4$. Then evaluate $f'(3)$.

$$f(x) = 3x^4, \quad x = 3$$

→ Power rule: $d/dx[3x^4] = 12x^3$.

$$\rightarrow f'(3) = 12(3)^3 = 324.$$

Answer: $f'(x) = 12x^3, \quad f'(3) = 324$



Product rule

24. Use the product rule to differentiate $f(x) = (3x + 4)(4x + 3)$. Find $f'(0)$.

$$f(x) = (3x + 4)(4x + 3), \quad x = 0$$

$$\rightarrow u = 3x+4, \quad v = 4x+3. \quad u' = 3, \quad v' = 4.$$

$$\rightarrow f'(x) = u'v + uv' = 3(4x+3) + 4(3x+4).$$

$$\rightarrow f'(0) = 3(3) + 4(4) = 25.$$

Answer: $f'(0) = 3 \cdot 3 + 4 \cdot 4 = 25$

25. Use the product rule to differentiate $f(x) = (1x + 3)(4x + 1)$. Find $f'(2)$.

$$f(x) = (1x + 3)(4x + 1), \quad x = 2$$

$$\rightarrow u = 1x+3, \quad v = 4x+1. \quad u' = 1, \quad v' = 4.$$

$$\rightarrow f'(x) = u'v + uv' = 1(4x+1) + 4(1x+3).$$

$$\rightarrow f'(2) = 1(9) + 4(5) = 29.$$

Answer: $f'(2) = 1 \cdot 9 + 4 \cdot 5 = 29$

26. Use the product rule to differentiate $f(x) = (2x + 4)(4x + 6)$. Find $f'(2)$.

$$f(x) = (2x + 4)(4x + 6), \quad x = 2$$

$$\rightarrow u = 2x+4, \quad v = 4x+6. \quad u' = 2, \quad v' = 4.$$

$$\rightarrow f'(x) = u'v + uv' = 2(4x+6) + 4(2x+4).$$

$$\rightarrow f'(2) = 2(14) + 4(8) = 60.$$

Answer: $f'(2) = 2 \cdot 14 + 4 \cdot 8 = 60$

27. Use the product rule to differentiate $f(x) = (1x + 3)(1x + 2)$. Find $f'(2)$.

$$f(x) = (1x + 3)(1x + 2), \quad x = 2$$

$$\rightarrow u = 1x+3, \quad v = 1x+2. \quad u' = 1, \quad v' = 1.$$

$$\rightarrow f'(x) = u'v + uv' = 1(1x+2) + 1(1x+3).$$

$$\rightarrow f'(2) = 1(4) + 1(5) = 9.$$

Answer: $f'(2) = 1 \cdot 4 + 1 \cdot 5 = 9$

28. Use the product rule to differentiate $f(x) = (4x + 2)(2x + 3)$. Find $f'(2)$.

$$f(x) = (4x + 2)(2x + 3), \quad x = 2$$

$$\rightarrow u = 4x+2, \quad v = 2x+3. \quad u' = 4, \quad v' = 2.$$

$$\rightarrow f'(x) = u'v + uv' = 4(2x+3) + 2(4x+2).$$

$$\rightarrow f'(2) = 4(7) + 2(10) = 48.$$

Answer: $f'(2) = 4 \cdot 7 + 2 \cdot 10 = 48$

29. Use the product rule to differentiate $f(x) = (4x + 3)(3x + 5)$. Find $f'(1)$.

$$f(x) = (4x + 3)(3x + 5), \quad x = 1$$

$$\rightarrow u = 4x+3, \quad v = 3x+5. \quad u' = 4, \quad v' = 3.$$

$$\rightarrow f'(x) = u'v + uv' = 4(3x+5) + 3(4x+3).$$

$$\rightarrow f'(1) = 4(8) + 3(7) = 53.$$

Answer: $f'(1) = 4 \cdot 8 + 3 \cdot 7 = 53$



30. Use the product rule to differentiate $f(x) = (1x + 1)(2x + 1)$. Find $f'(2)$.

$$f(x) = (1x + 1)(2x + 1), \quad x = 2$$

$$\rightarrow u = 1x+1, v = 2x+1. \quad u' = 1, v' = 2.$$

$$\rightarrow f'(x) = u'v + uv' = 1(2x+1) + 2(1x+1).$$

$$\rightarrow f'(2) = 1(5) + 2(3) = 11.$$

Answer: $f'(2) = 1 \cdot 5 + 2 \cdot 3 = 11$

